Manual Chapter : Configuring the BIG-IP System as a DHCP Relay Agent

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Configuring the BIG-IP System as a DHCP Relay Agent

Overview: Managing IP addresses for DHCP clients

When you want to manage Dynamic Host Configuration Protocol (DHCP) client IP addresses, you can configure the BIG-IP system to act as a DHCP relay agent. A common reason to configure the BIG-IP system as a DHCP relay agent is when the DHCP clients reside on a different subnet than the subnet of the DHCP servers.

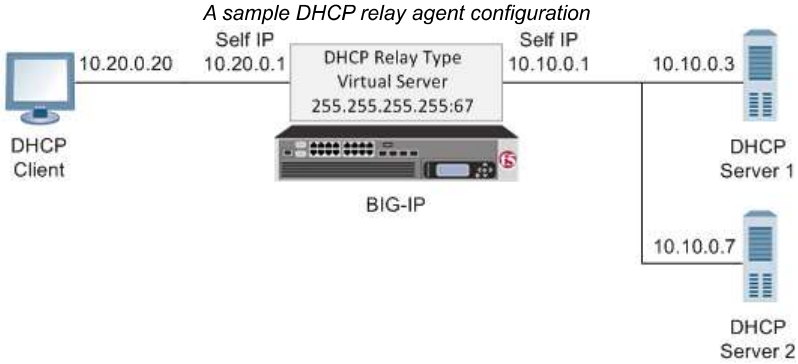
Before configuring the BIG-IP system to act as a DHCP relay agent, it is helpful to understand some BIG-IP system terminology:

BIG-IP object type	Definition
BIG-IP pool member	A DHCP relay target (such as a DHCP server or BOOTP server). This is the dynamic address server to which the BIG-IP system forwards unicast requests.
BIG-IP virtual server	A BIG-IP system address on the listening VLAN
BIG-IP VLAN assigned to a virtual server	A listening VLAN, controlled on a per-virtual server basis

About the BIG-IP system as a DHCP relay agent

A BIG-IP virtual server, configured as a Dynamic Host Configuration Protocol (DHCP) type, provides you with the ability to relay DHCP client requests for an IP address to one or more DHCP servers, available as pool members in a DHCP pool, on different +virtual local area networks (VLANs). The DHCP client request is relayed to all pool members, and the replies from all pool members are relayed back to the client.

A sample DHCP relay agent configuration



For example, a DHCP client sends a broadcast message to the destination IP address 255.255.255.255, which is the destination address configured on the virtual server. A DHCP type virtual server automatically uses port 67 for an IPv4 broadcast message or port 547 for an IPv6 broadcast message. The BIG-IP virtual server receives this message on the VLAN with self IP address 10.20.0.1 and relays the DHCP request to all DHCP servers: 10.10.0.3 and 10.10.0.7.

All DHCP servers provide a DHCP response with available IP addresses to the BIG-IP virtual server, which then relays all responses to the client. The client accepts and uses only one of the IP addresses received.

In this example, there is no hop between the DHCP client and the BIG-IP relay agent. However, a common topology is one that includes this hop, which is often another BIG-IP system.

Alternate configuration

If the DHCP client subnet includes a BIG-IP system that serves as a hop to the BIG-IP relay agent, you must perform two additional configuration tasks:

- You must configure the BIG-IP relay agent to relay the client DHCP requests to the DHCP servers without losing the originating subnet (source) IP address. This originating source IP address is typically a self IP address of the BIG-IP system that resides on the client subnet. You configure the BIG-IP relay agent to preserve the originating source IP address by creating a SNAT that specifies the originating self IP address as both the origin address and the translation address. A SNAT configured in this way prevents the BIG-IP relay agent, before sending the DHCP broadcast message to the DHCP servers, from translating the source IP address of the incoming DHCP request to a different address.
- You must add a route (to the BIG-IP relay agent) that specifies the originating source IP address as the destination for DHCP responses. The DHCP servers use this route to send their responses back through the BIG-IP relay agent to the clients.

Task summary for configuring system as a DHCP relay agent

You configure the BIG-IP system to act as a Dynamic Host Configuration Protocol (DHCP) relay agent by creating a pool of DHCP servers and then creating a virtual server to manage DHCP client broadcast messages.

Creating a pool of DHCP servers

You must create a pool that includes Dynamic Host Configuration Protocol (DHCP) servers as pool members before you create a DHCP type of virtual server.

Never add more than 20 pool members to a pool of DHCP servers. Although the BIG-IP system does not prevent you from adding more than 20 members to a pool, only 20 members or fewer will receive the client DHCP DISCOVER request.

1. On the Main tab, click **Local Traffic › Pools** .
   The Pool List screen opens.
2. Click **Create.**
   The New Pool screen opens.
3. In the **Name** field, type a unique name for the pool.
4. Type a description for the pool.
5. For the **Health Monitors** setting, in the **Available** list, select **UDP**, and click **<<** to move the monitor to the **Active** list.
6. From the **Load Balancing Method** list, select a method.

   A DHCP pool requires a load balancing method, although actual load balancing across DHCP pool members is ignored and DHCP requests are sent to all DHCP pool members.

7. For the **Priority Group Activation** setting, select **Disabled.**
8. Add each resource that you want to include in the pool using the **New Members** setting:
   a. Type a name in the **Node Name** field, or select a node address from the **Node List.**
   a. Type an IP address in the **Address** field, or select a node address from the **Node List.**
   b. Type 67 (IPv4) or 547 (IPv6) in the **Service Port** field.
   c. Click **Add.**
9. Click **Finished.**

A pool that includes DHCP servers as pool members is created.

Creating a DHCP type virtual server

A DHCP type of BIG-IP virtual server provides you with the ability to relay DHCP client requests for an IP address to one or more DHCP servers, and provide DHCP server responses with an available IP address for the client.

1. On the Main tab, click **Local Traffic › Virtual Servers** .
   The Virtual Server List screen opens.
2. Click **Create.**
   The New Virtual Server screen opens.
3. In the **Name** field, type a unique name for the virtual server.
4. Type a description for the virtual server.
5. From the **Type** list, select **DHCP.**
6. Select one of the following to configure a **Destination Address** type.

Destination	Steps to configure
255.255.255.255 (IPv4 Default)	None.
ff02::1:2 (IPv6 Default)	None.
Other	For a host or network, in the **Destination Address** field, type an IPv4 address/prefix or an IPv6 address/prefix.

7. From the **State** list, select **Enabled.**
8. In the Configuration area for the **VLAN and Tunnel Traffic** setting, select the VLANs on the same network as the DHCP clients to ensure that the BIG-IP system can accept the broadcast traffic from the client.
9. From the **Default Pool** list, select the pool that is configured for DHCP servers.
10. Click **Finished.**

A DHCP type of virtual server is configured to provide the ability to relay DHCP client requests for an IP address to one or more DHCP servers, and provide DHCP server responses with an available IP address for the client.

Implementation result

The BIG-IP system is configured to manage Dynamic Host Configuration Protocol (DHCP) client IP addresses, using a DHCP type of virtual server to manage DHCP client broadcast messages.

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